Cortical beta oscillations map to shared brain networks modulated by dopamine

Paper: Chikermane et al. (2024). Human recordings and network mapping.

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Page references below use PDF page numbers, including front matter.

To which general question does this paper relate?

Measurement and the network basis of beta activity. Useful for deciding what a beta measurement represents before selecting a feedback objective.

How does it answer the general questions?

Measured. Recordings selected from clinically normal cortical regions in epilepsy patients are combined with normative connectivity and dopamine-related PET maps. Regions with prominent beta activity share connectivity with a distributed cortico-subcortical network (PDF pp. 1–9).

Interpretation. Beta is also a physiological rhythm associated with several functions. The spatial correspondence with dopamine markers supports a network hypothesis; it does not demonstrate that beta causes symptoms or that suppressing it improves behaviour. This is not a DBS treatment trial.

Model relevance. Supports considering cortical inputs and distributed coupling rather than assigning beta exclusively to one nucleus. The connectivity maps do not identify dynamical coupling gains, transmission delays, or a firing-rate-to-LFP measurement equation for the notebook model (PDF pp. 8–12).

What papers does it cite that could provide more information?

- Hammond, Bergman and Brown (2007), *Pathological synchronization in Parkinson's disease: networks, models and treatments*: competing mechanisms (references, PDF p. 15).
- Gilron et al. (2021), *Long-term wireless streaming of neural recordings for circuit discovery and adaptive stimulation in individuals with Parkinson's disease*: chronic sensing (PDF p. 15).